Relevance-Gated Agentic Workflow for Autonomous Research Monitoring

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1 Introduction

The volume of research literature published every year, particularly in fast-moving areas of computer science such as machine learning and natural language processing, has grown to a point where it is no longer practical for an individual researcher or student to manually track every new paper relevant to their interests. Repositories such as arXiv add thousands of new submissions weekly, and while keyword search and category browsing help narrow the space, they place the entire burden of filtering and prioritization on the user. Most existing tools either surface too much irrelevant content or require constant manual re-tuning of search terms, and few provide any structured way to distinguish between papers that are clearly relevant, clearly irrelevant, and genuinely uncertain — nor do they offer any mechanism for the system to learn from how a user actually responds to what it surfaces.

This project addresses that gap by presenting a personalized research-monitoring system that autonomously discovers newly published papers and delivers a curated, scheduled digest tailored to an individual user's interests, rather than requiring the user to repeatedly issue search queries. Papers are represented as embeddings and compared against a user's interest profile, and the system's central design decision is to move beyond a single relevance score by treating relevance as a three-way routing problem: papers that are clearly relevant are processed and summarized automatically, papers that are clearly irrelevant are discarded (with a logged record rather than a silent drop), and papers whose relevance is genuinely uncertain are escalated to a human reviewer instead of being guessed at. This explicit routing is what makes the system agentic rather than a conventional retrieve-and-rank pipeline — the system makes an autonomous decision about how each paper should be handled, rather than simply producing a ranked list for the user to sift through.

The human-in-the-loop step is deliberately treated as more than a safeguard: a reviewer's accept or reject decision on a borderline paper is captured as a distinct signal alongside the end user's own like or dislike reaction to delivered papers, and both signals are stored as part of the user's preference memory. This preference memory is used to periodically retrain a lightweight logistic regression relevance classifier, allowing the system's filtering behaviour to adapt to the user's actual interests over time rather than relying on a fixed, hand-tuned rule. Retraining is performed in batches on accumulated feedback rather than after every single interaction, keeping the system's behaviour more stable and making its decisions traceable to a specific, inspectable model at each retraining cycle.

The overall objective of this project is to design and build this pipeline — discovery, relevance-gated routing, human review, automated summarization, scheduled delivery, and feedback-driven preference adaptation — as a working, deployable system evaluated with a real dataset of research papers and a small group of real users, so that its usefulness can be assessed through actual usage rather than illustrative examples alone. The system will be evaluated using precision and recall on a manually labelled validation subset, together with learning curves that examine filtering performance as accumulated feedback increases, and will be further assessed through a small-scale deployment with real users. The project deliberately favours a lightweight and explainable design over more complex alternatives such as fine-tuning, reinforcement learning, or heavy multi-agent orchestration, since the goal is a system whose every decision can be inspected and justified rather than one that is technically elaborate for its own sake.

The significance of this work lies not in any individual technique — embeddings, similarity search, and LLM-based summarization are all well-established — but in how they are integrated into a single coherent workflow that combines autonomous relevance gating, genuine human oversight of uncertain cases, and continual preference adaptation, applied specifically to the problem of ongoing research monitoring rather than one-off literature search.

2 Literature Review

Several existing systems address parts of the problem this project tackles, but the reviewed systems do not combine explicit relevance gating, human-in-the-loop review as a training signal, and lightweight periodic personalization in the way proposed here.

2.1 Personalized Paper Recommendation Systems

arXiv Sanity Preserver, developed by Karpathy (2021), was one of the earliest widely used tools for taming the volume of machine learning papers on arXiv. It represents papers using TF-IDF vectors over paper text and lets users build a personal library; a support vector machine is then trained per user on their library to rank new papers by similarity. This established the basic pattern of learning a per-user relevance model from lightweight signals, which the proposed system builds on. However, arXiv Sanity Preserver has no notion of an uncertain or borderline case — papers are simply ranked — and it does not incorporate any automated summarization or scheduled delivery step; the user must actively visit the site and browse.

A more recent large-scale example is the recommendation component of the Semantic Scholar platform, which trains a linear model (originally an SVM, extended from the arXiv Sanity approach) over embedding representations of papers a user has positively or negatively annotated, and uses approximate nearest-neighbour search to select candidate papers from a large, continuously refreshed index before ranking them. This is architecturally close to the classifier-based personalization used in the proposed system, but it operates as a ranking layer within a much larger literature search platform rather than as a self-contained monitoring agent, and it does not include an explicit human-review escalation path for uncertain cases.

2.2 Conversational and Agentic Literature Assistants

SURVEYAGENT is a conversational system designed to help researchers interact with scientific literature more efficiently, combining literature search with a dialogue interface and a degree of personalization. It demonstrates the value of moving beyond static ranked lists toward a more interactive, assistant-like experience. However, it is fundamentally query-driven: the user initiates each interaction, rather than the system autonomously monitoring a field and proactively surfacing new material on a schedule, which is the core behaviour the proposed system targets.

Separately, the broader body of work on Agentic Retrieval-Augmented Generation describes a shift from static retrieve-then-generate pipelines toward systems in which autonomous agents plan, use tools, and adapt their retrieval and reasoning strategies rather than following a fixed sequence. This literature provides the conceptual basis for what distinguishes an "agentic" system from a conventional RAG pipeline in this project: the presence of an autonomous routing decision (in this proposed system architecture's case, the three-way relevance gate) rather than a single fixed retrieve-and-answer step.

2.3 Human-in-the-Loop Agentic Systems

Closer in spirit to the proposed system's review mechanism is ORCHID, a demonstrated agentic system for high-risk property classification that pairs retrieval-augmented generation with human oversight, explicitly deferring uncertain classifications to subject-matter experts and logging the resulting decisions for auditability. This validates the general pattern of routing uncertain cases to a human reviewer rather than forcing an automated system to decide everything, which the proposed system adopts for its borderline-relevance path. ORCHID, however, is designed for a high-stakes compliance classification task with a multi-agent messaging architecture, which is considerably heavier than what is appropriate or necessary for a personal research-monitoring tool.

2.4 Identified Gap

Taken together, the reviewed systems each address part of the problem: personalized ranking (arXiv Sanity, Semantic Scholar), conversational access to literature (SURVEYAGENT), the general concept of autonomous agentic decision-making (the Agentic RAG literature), and human-in-the-loop escalation of uncertain cases (ORCHID). The reviewed systems do not combine an explicit three-way relevance gate, a human review step whose decisions feed directly back into the same learned preference model as end-user feedback, and a lightweight, periodically retrained, fully explainable classifier — all within a scheduled, autonomous monitoring workflow rather than a query-driven search tool. This combination is the specific gap the proposed framework is designed

to fill, with novelty arising from the integration rather than from any individual technique.

3 Design Figure

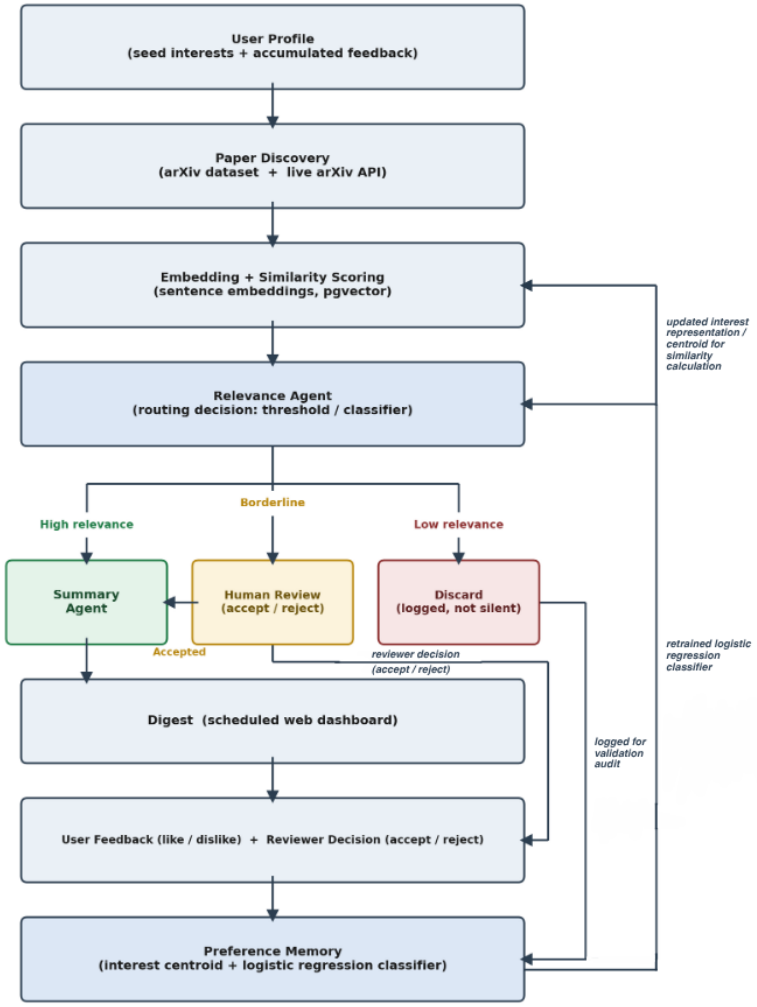


Figure 1: Proposed Research Monitoring System

Figure 1 shows the end-to-end architecture of the proposed system, implemented as an explicit Python state machine rather than through a multi-agent orchestration framework. The pipeline begins with the user's profile, which combines a small set of seed research interests with any feedback accumulated so far. Paper Discovery pulls candidate papers from both a static bulk dataset and a live arXiv API call, and each candidate is embedded and compared against the user's interest representation via cosine similarity in a pgvector-backed database.

The Relevance Agent is the central decision-making component: it routes each paper down one of three explicit paths based on a similarity threshold (during cold start) or a trained classifier (once sufficient feedback exists). High-relevance papers proceed automatically to the Summary Agent; borderline papers are escalated to a Human Review step where a reviewer makes an accept or reject decision, and accepted papers are likewise passed to the Summary Agent for processing, so that every paper reaching the digest — whether cleared automatically or through review — carries a consistently generated, interest-framed summary. Low-relevance papers are discarded, but this discard is logged rather than silently dropped, so it can later be audited against a validation set.

The reviewer's accept/reject decision and the end user's like/dislike reaction to delivered papers are captured as two distinct signals reaching Feedback Collection by two distinct paths: the reviewer's decision is recorded directly from the Human Review step regardless of outcome, while

end-user reactions are recorded from the digest the user actually receives. Both signals then flow into Preference Memory, which maintains an interest centroid and periodically retrains the logistic regression classifier used by the Relevance Agent. This closes the loop shown at the right of the diagram: each retraining cycle is intended to improve the quality of the Relevance Agent's routing decisions in the next cycle, which is the mechanism evaluated in Section 5.

4 Module Description

The proposed system comprises eight modules, summarized in Table 1 and described in more detail below.

Table 1: Modules of the Proposed Research Monitoring System

Module	Description
Paper Discovery	Pulls candidate papers from a static bulk dataset (arXiv metadata, filtered to relevant categories) and from a scheduled live call to the arXiv API.
Embedding + Similarity Scoring	Converts paper abstracts and the user's interest profile into vector embeddings and computes cosine similarity as the base relevance signal.
Relevance Agent	Makes the core autonomous routing decision for each paper: automatic processing, escalation to human review, or discard.
Summary Agent	Generates a short, interest-framed summary for each paper that clears the relevance gate, whether automatically or through human review.
Human Review	Presents borderline-relevance papers to a reviewer for an accept / reject decision.
Digest Delivery	Assembles all summarized, gate-cleared papers into a scheduled digest.
Feedback Collection	Captures end-user like / dislike reactions and reviewer accept / reject decisions.
Preference Memory	Maintains the interest centroid and periodically retrains the Relevance Agent's classifier.

4.1 Paper Discovery

Purpose: This module exists to supply the rest of the pipeline with a continuous stream of candidate papers, without requiring the user to manually search or browse. It is the entry point of the entire workflow.

Functionalities: It draws papers from two sources — a static bulk dataset of arXiv metadata restricted to a small set of relevant categories, used to bootstrap the system with a working corpus, and a scheduled call to the live arXiv API, used to pull newly published papers on an ongoing basis. Its inputs are the selected target categories and the monitoring configuration; its output is a stream of raw paper records (title, abstract, authors, category, identifier, date) passed forward for processing.

Interaction with other modules: Paper Discovery initiates each monitoring cycle using the configured categories and monitoring settings. Its output feeds directly into Embedding + Similarity Scoring, which is the next stage in the pipeline.

4.2 Embedding + Similarity Scoring

Purpose: This module translates raw paper text into a form that can be numerically compared against a user's interests, which is the foundation the Relevance Agent's routing decision depends on.

Functionalities: It generates a sentence embedding for each incoming paper's abstract and computes the cosine similarity between that embedding and the user's interest representation. Its input is the raw paper stream from Paper Discovery together with the user's current interest representation; its output is each paper annotated with a similarity score.

Interaction with other modules: It receives papers from Paper Discovery and reads the current interest representation from Preference Memory. Its scored output is passed to the Relevance Agent, which uses the similarity score during cold start and the classifier's relevance estimate once sufficient feedback has accumulated.

4.3 Relevance Agent

Purpose: This is the central decision-making component of the system and the primary source of its agentic behavior. Its purpose is to autonomously decide how each paper should be handled, rather than presenting the user with an undifferentiated ranked list.

Functionalities: For each scored paper, it applies a routing rule. During the cold-start phase, routing is based on predefined similarity thresholds. Once sufficient feedback has accumulated, the system uses a periodically retrained logistic regression classifier, with the paper embedding as its input, to estimate relevance. Based on the resulting score or relevance estimate, the agent assigns one of three routing labels: high relevance, borderline relevance, or low relevance.

Interaction with other modules: It receives scored papers from Embedding + Similarity Scoring and, when available, the current classifier from Preference Memory. Based on the routing label, it forwards high-relevance papers to the Summary Agent, borderline papers to Human Review, and low-relevance papers to the Discard path. It is also the module whose relevance assessment is adapted through the feedback loop, since Preference Memory periodically retrains the classifier it uses.

4.4 Summary Agent

Purpose: This module exists to reduce the effort required for the user to judge a paper's relevance once it has already cleared the relevance gate, by producing a summary specifically framed around why the paper matches the user's interests rather than a generic abstract restatement.

Functionalities: It takes a gate-cleared paper's abstract and the user's stated interests as input and produces a short natural-language summary as output, generated through a single call to an LLM API per paper.

Interaction with other modules: It receives high-relevance papers directly from the Relevance Agent, and it also receives borderline papers that a human reviewer has accepted, ensuring that every paper reaching the digest — regardless of which path it took through the relevance gate — is summarized consistently. It passes its generated summaries forward to Digest Delivery.

4.5 Human Review

Purpose: This module is the system's explicit human-in-the-loop safeguard, ensuring that papers whose relevance is genuinely ambiguous are not silently mishandled by an automated decision. It also serves as a training-data source for the system's feedback loop.

Functionalities: It presents each borderline paper to a human reviewer through a simple interface, and records the reviewer's accept or reject decision. Its input is the set of borderline papers flagged by the Relevance Agent; its output is the reviewer's decision, associated with the corresponding paper.

Interaction with other modules: It receives borderline papers from the Relevance Agent. Accepted papers are passed to the Summary Agent for processing before inclusion in the digest, while the accept/reject decision itself is additionally sent to Feedback Collection, where it is treated as a distinct signal from end-user feedback but is combined with it in Preference Memory.

4.6 Digest Delivery

Purpose: This module exists to deliver curated results to the user on a recurring schedule, which is what distinguishes the proposed system as a monitoring system rather than a query-driven search tool.

Functionalities: It assembles all papers that have cleared the relevance gate — including automatically processed papers and papers accepted through Human Review — into a scheduled digest, with each paper's summary included, since every gate-cleared paper is routed through the Summary Agent regardless of which path it took.

Interaction with other modules: It receives summarized papers from the Summary Agent, which is now the sole source of summaries for both automatically processed and human-accepted papers, and presents them to the user. The digest it delivers is what the user reacts to in Feedback

Collection, making it the direct source of the end-user feedback signal.

4.7 Feedback Collection

Purpose: This module exists to capture the signals the system needs in order to adapt its future relevance decisions to the individual user, rather than relying on a static, hand-tuned rule indefinitely.

Functionalities: It records two distinct types of signal: end-user like/dislike reactions to papers delivered in the digest, and the human reviewer's accept/reject decisions on borderline papers. Each recorded feedback event is stored together with a reference to the corresponding paper and user profile, keeping the two signal types clearly distinguishable rather than merged into one undifferentiated feedback pool.

Interaction with other modules: It receives end-user reactions from Digest Delivery and reviewer decisions from Human Review. Its combined output — the full set of labeled feedback events — is passed to Preference Memory, where it is used for retraining.

4.8 Preference Memory

Purpose: This module closes the feedback loop, allowing the system's filtering behavior to adapt to the individual user's actual interests rather than remaining fixed after initial setup.

Functionalities: It maintains the user's evolving interest representation through an interest centroid that incorporates both positive feedback (likes, reviewer acceptances) and negative feedback (dislikes, reviewer rejections), and periodically retrains the logistic regression classifier used by the Relevance Agent on the accumulated feedback events. Retraining is performed in batches once a sufficient number of new feedback points have been collected, rather than continuously.

Interaction with other modules: It receives labeled feedback events from Feedback Collection, which include both end-user and human-reviewer signals. Its output — the updated interest centroid and retrained classifier — is read by the Relevance Agent and by Embedding + Similarity Scoring in subsequent monitoring cycles, which enables the system to evaluate whether its routing decisions improve over successive cycles of use.

5 Dataset and Method Description

5.1 Dataset Description

The project uses the publicly available arXiv metadata dataset (sourced via Kaggle/HuggingFace mirrors of Cornell University's arXiv data), which contains structured metadata — including title, abstract, authors, subject categories, submission and update dates, and a unique identifier — for the full corpus of arXiv submissions.

Rather than working with the entire multi-million-record corpus, the project restricts itself to a working subset of approximately 50,000–100,000 records drawn from two to three subject categories most relevant to the pilot users (for example cs.CL, cs.LG, and cs.AI), which keeps embedding generation and storage costs reasonable while still providing a realistic volume of data for evaluation.

Preprocessing steps applied to the raw metadata include: deduplication of records, filtering to the selected categories and a bounded recent date range, basic text cleaning of titles and abstracts (whitespace and encoding normalization), and batch generation of sentence embeddings for each abstract, which are stored in a pgvector-enabled Postgres table alongside the paper's metadata for efficient similarity search.

5.2 Method Description

Relevance scoring is based on cosine similarity between a paper's abstract embedding and the user's interest profile embedding, initially derived from a small set of user-provided interest descriptions and later updated using an interest centroid computed from liked and reviewer-accepted papers, with the centroid adjusted away from disliked and reviewer-rejected papers. This threshold-based routing is used during the cold-start phase, before sufficient feedback has accumulated. Papers that fall below the low-relevance threshold are discarded, but the discard decision is logged together with the paper's score rather than dropped silently, allowing discarded papers to be audited against a labeled validation set to check whether relevant papers were incorrectly filtered out.

Once approximately 30–50 feedback points (combining end-user reactions and human reviewer

decisions) have been collected, a logistic regression classifier is trained with the paper embedding as input and the observed relevant/not-relevant label as the target. This model was chosen deliberately over more complex alternatives such as fine-tuning a language model or applying reinforcement learning: logistic regression is fast to retrain, fully explainable (its coefficients can be inspected directly), and appropriate for the relatively small amount of feedback data realistically obtainable from a handful of pilot users within the project's evaluation window. The classifier is retrained periodically in batches rather than after every individual feedback event, which keeps the system's behaviour stable and avoids the cost of continuous retraining.

Summarization is performed via a single call per gate-cleared paper to a hosted LLM API, using a prompt that conditions the summary on the user's stated interests so that the output explains why the paper is relevant rather than restating the abstract. The overall workflow — discovery, scoring, routing, summarization, review, delivery, and feedback-driven retraining — is implemented as an explicit sequence of Python functions operating on a shared state object, rather than through a general-purpose multi-agent framework, so that every routing decision in the pipeline remains inspectable and traceable for evaluation.

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